



Nicotinamide adenine dinucleotide emerges as a therapeutic target in aging and ischemic conditions

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Abstract Nicotinamide adenine dinucleotide (NAD⁺) has been described as central coenzyme of redox reactions and is a key regulator of stress resistance and longevity. Aging is a multifactorial and irreversible process that is characterized by a gradual diminution in physiological functions in an organism over time, leading to development of age-associated pathologies and eventually increasing the probability of death. Ischemia is the lack of nutritive blood flow that causes damage and mortality that mostly occurs in various organs during aging. During the process of aging and related ischemic conditions, NAD⁺ levels decline and lead to nuclear and mitochondrial dysfunctions, resulting in age-related

pathologies. The majority of studies have shown that restoring of NAD⁺ using supplementation with intermediates such as nicotinamide mononucleotide and nicotinamide riboside can be a valuable strategy for recovery of ischemic injury and age-associated defects. This review summarizes the molecular mechanisms responsible for the reduction in NAD⁺ levels during ischemic disorders and aging, as well as a particular focus is given to the recent progress in the understanding of NAD⁺ precursor's effects on aging and ischemia.

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Introduction

Aging is a multifactorial biological process, causing disturbances in homeostasis, reduction in physiological function and immune responses as well as increased vulnerability to stressful insults or damages (Imai and Guarente 2014; López-Otín et al. 2013). The time-dependent accumulation of cellular damage in the body is considered as the general reason for aging (Gems and Partridge 2013). Senescence in human's cells starts after the third decade, which is accompanied by a decrease in wellness, mobility and the onset of various age-related diseases, including ischemic heart disease, hypertension, atherosclerosis, and neurodegenerative disorders (Liu 2014). The free radical and mitochondrial theories are the two most important biochemical theories that explain aging pathology. The free-radical theory proposed by Harman in 1956, states that aging is a consequence of free radicals attack to the cells and tissues (Harman 1956). However, mitochondrial theories suggests that accumulation of damage to mitochondria and mitochondrial DNA (mtDNA) is the main cause of aging (Miquel et al. 1980).

Nicotinamide adenine dinucleotide (NAD^+) is well-known as a cofactor in many biological functions. In addition to its important effect in energy metabolism, NAD^+ also has a critical role in calcium homeostasis, gene expression, signal transduction, and immunological functions and can delay aging and cell death (Houtkooper and Auwerx 2012). It is a substrate for several NAD^+ -dependent enzymes, for example, sirtuins, ADP-ribosyl cyclase (CD38), poly (ADP-ribose) polymerase 1 (PARP-1) and NAD^+ -dependent dehydrogenases (Liu et al. 2008). The level of this cofactor generally reduces during ischemic insults (Yamamoto et al. 2014) and aging process (Nakagawa and Guarente 2014), because of an increment in the activity of NAD^+ -consuming enzymes (NADases), and a reduction in NAD^+ -synthesizing enzymes such as nicotinamide phosphoribosyltransferase (NAMPT) and nicotinic acid phosphoribosyltransferase (NAPRT) (Scheibye-Knudsen et al. 2014). Moreover,

NAD^+ and its reduced form (NADH) are essential cofactors of a number of oxidoreductases. In a number of redox reactions in energy metabolism, NAD^+ was consumed by several dehydrogenases that catalyze substrate oxidation (Ying 2008). NAD(P)H quinone oxidoreductase-1 (NQO1) is a cytosolic flavoprotein that can modulate the cellular NAD^+/NADH ratio. The enzymatic function of NQO1 is reduced in aged tissues (Lee et al. 2012). In addition, cytochrome-b5 reductase, known as methemoglobin reductase, is a NADH-dependent enzyme and its activity decreases with increasing age, leading to reduced NAD^+/NADH ratio (Bueno 2018).

Ischemia abolishes the nutrient accessibility and oxygen delivery to the tissue and increases the accumulation of waste metabolites in the tissue. Here, the restoration of blood flow to the tissue, or reperfusion, has been established as the primary treatment to prevent additional tissue injury. Apoptosis, inflammation, oxidative damage, reactive oxygen species (ROS) overproduction, endothelial dysfunction, and cytosolic and mitochondrial Ca^{+2} overload are considered as the underlying mechanisms of ischemia-reperfusion (I/R) injury (Murphy and Steenbergen 2008; Song et al. 2014). Aging is a major risk factor for ischemic disorders. Previous studies have indicated that aging increases the susceptibility of the heart and various organs (e.g. brain, liver, lung, and kidney) to I/R injury (D'Annunzio et al. 2016; Moraga et al. 2015). Moreover, functional recovery after I/R injury is impaired in the elderly. Indeed, there is a link among aging, mitochondrial dysfunction, and increased susceptibility to I/R injury. Since the mitochondria are the most sensitive target for damage during I/R injury, and the other hand, the levels of NAD^+ are significantly reduced by aging and ischemic states, it is conceivable that NAD^+ -dependent enzymes activity and thereby the mitochondrial function would decrease in those situations (Lee et al. 2016).

A number of studies have demonstrated that boosting intracellular NAD^+ levels through supplementation with exogenous NAD^+ and its precursors have favorable effects in postponing the aging phenotypes and protect against I/R injury in various tissues through promoting mitochondrial function, activating autophagic flux, increasing antioxidant capacity and reducing oxidative stress and apoptosis (de Picciotto et al. 2016; Xie et al. 2017a; Zhang et al. 2016). Therefore, restoring the NAD^+ levels and

replenishment of its cellular pools to prevent its depletion can be considered as a therapeutic strategy for prevention and treatment of aging-associated diseases including ischemic disorders. Therefore, in this review, we attempt to explain the effect of NAD^+ on ischemic disorders and aging process and discuss up-to-date and future opportunities for NAD^+ -based therapeutic approach for ameliorating ischemia and aging-related dysfunctions.

NAD^+ biosynthesis

NAD^+ was firstly discovered over 110 years ago as a low molecular weight substance present in a boiled yeast extract (Harden and Young 1906). Hans von Euler-Chelpin has revealed that NAD^+ is composed of two mononucleotides adenosine monophosphate (AMP) and nicotinamide mononucleotide (NMN) (von Euler 1999). NADH , the reduced form of NAD^+ , is a hydrogen-donor in the aerobic production of ATP (Wallace 2012). Cytosolic levels of NAD^+ are higher than nuclear levels (Zhang et al. 2009) and lower than mitochondrial levels ($\geq 250 \mu\text{M}$) (Yang et al. 2007). It is estimated that intracellular NAD^+ has a short half-life between 1 to 2 h (Elliott and Rechsteiner 1975; Rechsteiner et al. 1976). The mammalian NAD^+ can be synthesized from itself and five major substrates: tryptophan (Trp), nicotinic acid (NA), nicotinamide riboside (NR), NMN and nicotinamide (NAM). In cells, NAD^+ is synthesized from three main pathways: (1) from Trp through the de novo biosynthesis pathway (L-tryptophan/kynurenine pathway), (2) from NA in the Preiss–Handler pathway; and (3) from NAM, NR, and NMN in the salvage pathway (Fig. 1).

De novo biosynthesis pathway

De novo biosynthesis of NAD^+ begins with metabolism of L-Trp via a multi-step oxidation reaction process to quinolinic acid (QA). Subsequently, QA is decarboxylated by quinolinate phosphoribosyltransferase (QPRT) to nicotinic acid mononucleotide (NAMN). Ultimately, NAMN is converted into NAD^+ through the Preiss–Handler pathway. The uptake of L-Trp and consequently the metabolic

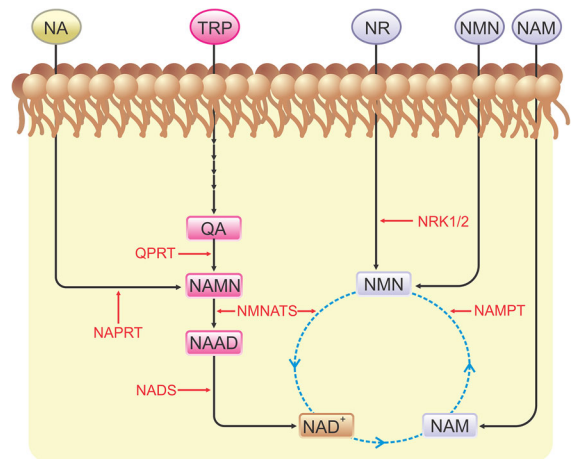


Fig. 1 Schematic representation of pathways for NAD^+ biosynthesis. NAD^+ can be synthesized by Preiss–Handler, de novo, and salvage pathways (from left to right). Multiple enzymes such as CD38, Sirtuins and PARPS consume NAD^+ and produce nicotinamide. QA Quinolinic acid, QPRT Quinolinate phosphoribosyltransferase, TRP Tryptophan, NA Nicotinic acid, NAMN Nicotinic acid mononucleotide, NADS NAD synthase, NMN Nicotinamide mononucleotide, NRK1/2 Nicotinamide riboside kinase, NAMPT Nicotinamide phosphoribosyltransferase, NR Nicotinamide riboside, NAM Nicotinamide. All pathways have been explained in the text

efficiency of de novo biosynthesis of NAD^+ is reduced in senescence (Braidys et al. 2011; Levine et al. 2014).

Preiss–Handler pathway

In this pathway, NA (niacin or vitamin B3) is first metabolized to NAMN by nicotinic acid phosphoribosyltransferase (NAPRT) enzyme. After that, nicotinamide mononucleotide adenylyltransferase (NMNATs) converts NAMN into NA adenine dinucleotide (NAAD), which is converted to NAD^+ by the action of NAD synthase (NADS) enzyme.

Salvage pathway

Nicotinamide riboside (NR) and NAM are key NAD^+ precursors that are converted into NMN by NRKs and NAMPT enzymes, respectively. Thereafter, NMN is adenylated to NAD^+ by the action of

NMNATs. NAMPT, or visfatin, is the rate-limiting enzyme in NAD^+ synthesis and influences the activity of NAD -dependent enzymes Sirtuins and PARPs, and thereby regulates mitochondrial biogenesis, cellular metabolism and inflammatory responses (Rodgers et al. 2008; Van Gool et al. 2009). The studies show that the levels of both NAMPT and NMNAT enzymes decline in aging (Stein and Imai 2014) as well as ischemia (Hsu et al. 2009). NAMPT is found in most cells and tissues and has two intra- and extracellular forms (iNAMPT and eNAMPT, respectively) in mammals (Samal et al. 1994). The iNAMPT is involved in intracellular NAD^+ synthesis (Kitani et al. 2003). However, eNAMPT is a cytokine released from neutrophils, mesangial cells, cardiomyocytes, and adipocytes and is responsible for cellular NMN synthesis (Pillai et al. 2012; Revollo et al. 2007; Song et al. 2008).

NAD^+ levels decline with aging process and ischemic injury

In addition to a reduction in the levels of NAD^+ biosynthesis enzymes, activation of PARP-1 and CD38 (a NAD^+ hydrolase) as well as overproduction of oxidative stress, and inflammation are responsible for a decrease in NAD^+ levels during aging and ischemic insults (Fig. 2). PARP-1 is a NADase enzyme involved in DNA repair and programmed cell death (Veith and Mangerich 2015). Free radicals produced during aging and reperfusion injury induce DNA damage and activate PARP1 enzyme, resulting in depletion of intracellular NAD^+ and ATP, as well as the reduction in SIRT1 activity and cell death (Baxter et al. 2014; Becatti et al. 2012; Di Lisa et al. 2000). Increased poly-ADP-ribose levels has been demonstrated after 2 h of recovery following ischemic injury that is attributed to NAD^+ depletion (Park et al. 2016; Strosznajder et al. 2003).

The NAD^+ consuming enzyme CD38 regulates cell proliferation, hormone release, inflammatory response, and muscle contraction (Chini 2009). It has been reported that CD38 levels and its activity are increased during the course of aging, leading to an augmented consumption of NAD^+ (Polzonetti et al. 2012). The increased CD38 activity is related to the reduction of mitochondrial NAD^+ levels and cause mitochondrial dysfunction through a pathway

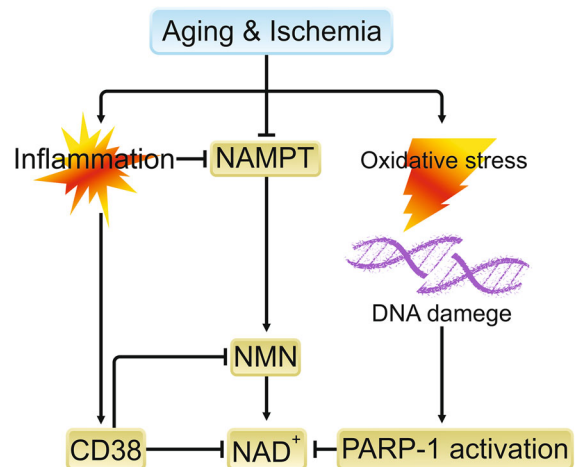


Fig. 2 The molecular mechanisms that are involved in the reduction of NAD^+ levels during aging and ischemia. Aging and ischemic injury lead to increased oxidative stress via reactive oxygen species overproduction, DNA damage and inflammation which decrease the levels of NAD^+ through activation of CD38 and PARP-1. NAMPT expression reduces with advance age and in ischemic conditions that cause reduced production of NMN and NAD^+

mediated, at least in part, by SIRT3 activity (Claire et al. 2003). The study by Chini et al. has shown that over-expression of CD38 in cancer cells contributed to increased features of cellular senescence (Chini et al. 2014). Moreover, a study in CD38 knock-out mice (CD38KO) disclosed that the tissue levels of NAD^+ in wild-type mice were lower than in CD38KO animals. So, it can be claimed that CD38 is a critical regulator of cellular NAD^+ levels during aging (Aksoy et al. 2006).

CD38 is activated following cerebral ischemia and results in a reduction of NAD^+ storage in brain tissue. CD38KO animals demonstrate protective effects against brain ischemic injury through the reduction in migration and accumulation of T-cells and macrophages in ischemic areas (Choe et al. 2011; Long et al. 2017). In addition, CD38 deficiency also protects the heart from I/R injury by activating SIRT1/FOXOs-mediated anti-oxidative pathway in myocardial cells (Guan et al. 2016). Moreover, CD38 is known as one of the main NMN-degrading enzymes that metabolizes NAD^+ and NR and decreases the accessibility of NAD^+ precursors to cells during the aged process and ischemia (Camacho-Pereira et al. 2016). Therefore, the combination of NAD^+ replacement-based therapies with inhibitors of CD38 can be useful to achieve

the protective effects against ischemic damage and aging.

Chronic inflammation and oxidative stress are other pathological conditions related to the aging process and ischemic insults. Chronic inflammation reduces the NAMPT expression and increases the CD38 expression and causes NAD⁺ depletion (Imai and Yoshino 2013). A number of studies have revealed the complex link between NAMPT and chronic inflammation. These studies stated that NAMPT expression can be either reduced or increased by inflammatory cytokines (Hsu et al. 2009; Yoshino et al. 2011) and oxidative stress leads to the excessive activation of PARP-1 which in turn reduces NAD⁺ levels and thereby mitochondrial membrane potential (Chen et al. 2011).

Effects of NAD⁺ supplementation in aging

Recent studies have shed light on a prominent role of NAD⁺ and its precursors in lifespan extension (Mouchiroud et al. 2013) and treatment of age-associated diseases (Cantó et al. 2012; Fang et al. 2014; Scheibye-Knudsen et al. 2014) (Fig. 3). Mills and colleagues observed that long-term administration of NMN improved many age-associated disorders in an animal model. They reported that NMN improved mitochondrial energy metabolism, insulin sensitivity, plasma lipid profile, eye function, physical activity, and body weight gain (Mills et al. 2016). In another study on aged mice, NMN has restored the arterial SIRT1 activity and improved the age-related endothelial dysfunction. In addition, their findings elucidated that NMN supplementation significantly restored NAD⁺ levels and decreased oxidative stress (de Picciotto et al. 2016). Along with these studies, Guan et al. have investigated the effect of NMN supplementation on renal dysfunction in an aged rodent model and reported that it restored the NAD⁺ content and SIRT1 activity and consequently protected the kidney from acute kidney injury (Guan et al. 2017). Additionally, the NAD⁺ levels, NAMPT expression and neural stem cells (NSCs) pool were reduced in the hippocampus of aged C57Bl6 mice. Long-term NMN administration in these animals augmented the NAD⁺ level and maintained the NSCs pool in their dentate gyrus (Stein and Imai 2014). Moreover, hearing loss in older animals has been reversed by boosting NAD⁺

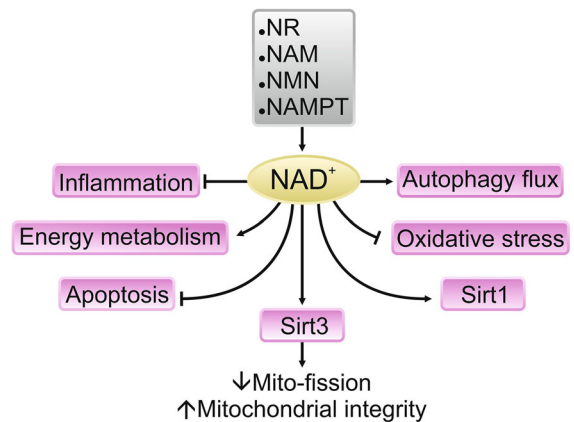


Fig. 3 Schematic diagram shows NAD⁺ targets which are involved in reduction of ischemic injury and aging pathology. The reduced levels of NAD⁺ in aging and ischemia are reversed by the use of NR, NAM, NMN or increased expression of NAMPT. Oxidative stress, inflammation, and apoptosis are induced by aging and ischemic insult. NAD⁺ can inhibit these processes. NAD⁺ also plays a protective role by increasing autophagy flux, energy metabolism and activation Sirtuin 1 and 3 (Sirt1& 3). Higher SIRT3 activity can inhibit mitochondrial fission (Mito-fission) and increase mitochondria integrity

levels and activating SIRT3 (Brown et al. 2014). Also, when NMN supplementation used in age-diabetic mice, it improved impaired glucose tolerance and lipid profiles (Yoshino et al. 2011). In addition, a study showed that NADH treatment (10 days, once a day, intraperitoneal injection) in old Wistar rats counteracted learning deficits and improved water maze performance as the indicator of cognition (Rex et al. 2004). The effects of NAD⁺ on aging have been summarized in Table 1. Collectively, these studies illustrate that restoration of NAD⁺ levels using its supplementation can be a good therapeutic target for preventing or postponing the aging phenotypes.

The effect of NAD⁺ restoration upon ischemia and reperfusion injury

Heart

Ischemic heart diseases are one of the main leading causes of mortality, disability, and lower quality of life among the elderly population in the world (Khalili et al. 2012). Myocardial ischemia is due to complete or partial obstruction of one or more branches of coronary arteries and ultimately cause cardiomyocytes

Table 1 The effects of NAD⁺ modulation in aging animal models

NAD ⁺ — treatments	Dose/time	Species	Organ	Main findings	References
NMN	100 or 300 mg/kg/day for 1 year	Aged mice	Eye, skeletal muscle	↓Weight gain, improved insulin sensitivity and physical activity, better eye function	Mills et al. (2016)
NMN	300 mg/kg/day 2 months	Aged mice	Arterial	Restoring NAD ⁺ and NO bioavailability and SIRT1 activity, ↓oxidative stress	de Picciotto et al. (2016)
NMN	500 mg/kg/day, 4 days	Aged mice	Renal	↑NAD ⁺ and SIRT1 activity	Guan et al. (2017)
NMN		Mice	Hippocampal	↑NAD ⁺ level	Stein and Imai (2014)
NR	Twice daily (1000 mg/kg) for 5 days prior to noise	Mice	Ear	↑NAD ⁺ level, activation SIRT3	Brown et al. (2014)
NMN	500 mg/kg 1 week	Aged mice		Improvement impaired glucose tolerance and lipid profiles	Yoshino et al. (2011)
NMN NR	500 mg/kg for 1 week	Aged mice	skeletal muscle	Better mitochondrial function	(Frederick et al. 2016; Mouchiroud et al. 2013)

death (Bayrami et al. 2018). Reperfusion following ischemia or hypoxia reduces the myocardial infarction (MI) size, preserves left ventricle systolic function and prevents the development of heart failure. However, reperfusion itself is accompanied by aggravated myocardial damage, arrhythmias, microvascular injury, myocardial dysfunction, and myocardial stunning and no-reflow phenomenon (Badalzadeh et al. 2017; Saeid et al. 2018). Zhang et al. examined the role of NAD⁺ in myocardial I/R injury in a rat model. They found that intravenous administration of NAD⁺ can significantly decrease the myocardial ischemic injury by reducing apoptotic damage and enhancing the antioxidant capacity of the heart (Zhang et al. 2016). Moreover, another study revealed that NAD⁺ precursor, NMN exerts significant cardioprotective effect against I/R injury through decreasing FOXO1 acetylation and increasing SIRT1 activity besides stimulation of autophagy flux (Yamamoto et al. 2014). Similarly, it has been suggested that exogenous NAD⁺ is cardioprotective against hypoxia-induced myocardial injury by attenuation of apoptosis through restoring SIRT1 activity and inhibiting P53 activity (Liu et al. 2014). NR, a natural NAD⁺ precursor present in milk, can alleviate the myocardial I/R injury by improving the mitochondrial biogenesis and regulating autophagy through SIRT3-PGC-1 α /P53

pathway (Ping et al. 2016). Moreover, nicotinamide-rich diet increased the heart resistance to ischemic damage by up-regulation of sulfonylurea receptor (SUR2A) which is a regulatory subunit of sarcolemmal ATP-sensitive potassium channel. Also, Transgenic mice that solely overexpressed SUR2A represent decreased myocardial infarct size and increased cardiac resistance to I/R injury (Sukhodub et al. 2010). It has been reported that 10 mmol/l nicotinamide pretreatment decreased hypoxia-induced cardiomyocytes cell necrosis and apoptosis by improving mitochondrial stress, including a reduction in mitochondrial ROS levels through upregulation of antioxidant enzymes and prevention of mitochondrial membrane potential collapse (Tong et al. 2012).

The expression of NAMPT, a critical gatekeeper of energy status and survival in cardiac myocytes, decreases in the heart following myocardial I/R injury. Interestingly, overexpression of NAMPT in the heart of transgenic mice enhanced NAD⁺ content, diminished myocardial infarct size and apoptosis and also stimulated the autophagy flux through activation of SIRT1 (Hsu et al. 2009). It has been found that administration of eNAMPT decreases myocardial reperfusion-induced injury by activation of PI3 K-Akt and MEK1/2-Erk1/2 pathways and inhibition of mPTP opening (Lim et al. 2008). Xiao et al. found that

pretreatment with eNAMPT improved cell viability and attenuated mitochondrial membrane potential depolarization in response to H₂O₂ insult. It also inhibited the mitochondria-dependent apoptotic pathway through regulation of P53 and Bcl-2 family genes via AMPK activation (Xiao et al. 2013). Taken together, these data indicate that NAD⁺ has the capacity to activate SIRT1 and SIRT3 in the heart and improve cardiac functions. Further studies will identify the other mechanisms mediating the therapeutic benefits of NAD⁺ supplementation in cardiac I/R injury.

Brain

Brain stroke caused by ischemia is a major reason for death and the second chronic cause of disability during aging in humans around the world (Romero 2007). This type of stroke occurs when a thrombus or embolus blocks cerebral blood flow, usually in the middle cerebral artery, and consequently leads to the neuronal damage and death (Salehpour et al. 2019). It has been suggested that maintaining the cellular NAD⁺ levels via administration of either NAD⁺ (Ying et al. 2007) or PARP inhibitors (Matsuura et al. 2011) alleviate ischemic damages of the brain. Intranasal administration of NAD⁺ after reperfusion have reduced the neurological deficits and infarct formation (Ying et al. 2007). Moreover, administration of NMN subsequent to transient forebrain ischemia has a protective effect against the death of CA1 pyramidal neurons (Park et al. 2016). An study has shown that nicotinamide injection up to 2 h after the onset of transient focal cerebral ischemia reduced the infarct volume and improved both motor and sensory behavioral deficits in rats (Mokudai et al. 2000). It has been found that NAM promotes remyelination after stroke through activating NAD⁺/BDNF/TrkB pathway (Wang et al. 2017). Bi et al. reported that administration of NAM and NAD⁺ after ischemia reduces the neuronal death and increases the mitochondrial biogenesis in neurons (Bi et al. 2012). Moreover, delayed NMN administration (with the first dose at 12 h post-ischemia) was unable to protect against neuronal deficits and to reduce brain infarction. Nevertheless, it improved regenerative neurogenesis after brain ischemia (Zhao et al. 2015). Wei and coworkers reported that NMN improves tissue plasminogen activator (tPA)-induced hemorrhagic

transformation (HT) after cerebral ischemia through maintaining the integrity of the blood–brain barrier (BBB) in mice (Wei et al. 2017).

FK866, a highly-specific NAMPT inhibitor, aggravates brain infarction in ischemic stroke, whereas up-regulation of NAMPT protects neurons from ischemic brain damage by the SIRT1-dependent AMPK pathway. SIRT1 is co-expressed with serine/threonine kinase 11 (LKB1) which is an upstream kinase of AMPK and promotes LKB1 deacetylation in neurons (Wang et al. 2011). Interestingly, Zhao and coworkers showed that NAMPT overexpression promotes neurogenesis, increases survival rate and accelerates neurological functional recovery and body weight gain after cerebral ischemia (Zhao et al. 2015). A study has reported that blood NAMPT concentration is increased during ischemic stroke in human (Lu et al. 2009). Similarly, plasma concentrations of NAMPT, NAD⁺, and ATP were increased after the induction of brain ischemia in mice (Zhao et al. 2014). NAMPT secretion from glial cell enhances under oxygen–glucose deprivation (OGD) stress conditions. Moreover, NAMPT administration significantly diminishes OGD-induced cell death and apoptosis in both neurons and glial cells. In addition, plasma concentrations of NAMPT were reduced in 6-month-old but not 3-month-old spontaneously hypertensive stroke-prone (SHR-SP) rats (a spontaneous stroke animal model). The reduction in plasma levels of NAMPT is associated with susceptibility to stroke at around 6-months of age. These findings show that eNAMPT has a neuroprotective role in ischemic stroke (Zhao et al. 2014).

Autophagy is activated during induction of cerebral I/R injury in the adult rodents (Smith et al. 2011). The study by Zheng et al. reveals that elevated autophagy contributes to brain ischemic damage in mice model of transient middle cerebral artery (tMCA) occlusion, whereas NAD⁺ administration attenuates autophagy-related deficits and reduces infarct size, edema formation and neurological impairments induced by cerebral ischemia (Zheng et al. 2012). In contrast, the data from the study of Wang et al. indicates that NAMPT induces autophagy and thereby promotes neuronal survival during cerebral ischemia. In this respect, 3-methyladenine, an autophagy inhibitor, could partially abrogate the neuroprotective effects of NAMPT (Wang et al. 2012). Thus, it is still controversial whether activation of autophagy is a

neuroprotective phenomenon or a detrimental process during brain ischemia. Moreover, these studies and other reports indicate that NAD⁺ treatment has diverse effects on autophagy. While NAD⁺ reduces the cellular injury and the autophagic process in primary cell cultures or brain tissues under ischemic conditions (Alano et al. 2004, 2010), it can also induce autophagy of brain tumor cells (Han et al. 2011). This disparity is more likely explained by the dosages of NAD⁺ used in different studies, the types of cells as well as other factors which to be determined. Future studies can delineate the mechanisms underlying the different roles of NAD⁺ on autophagy in brain cells.

Mitochondrial quality control is mediated by mitophagy and mitochondrial biogenesis. Reports on the activation of mitophagy during aging and ischemic insults as well as the effect of NAD⁺ supplementation on this process are also controversial. During the reperfusion phase of I/R injury, the elevated ROS levels can induce over-activation of mitophagy (Li et al. 2012). Nevertheless, during senescence, the removal of damaged mitochondria via mitophagy is diminished, leading to a decrease in normal mitochondrial content and an increase in nonfunctional mitochondrial mass. However, NR and NAM can induce mitophagy in brain tissue (Jang et al. 2012). In this line, NAD⁺/SIRT1 has been shown to stimulate mitophagy through the forkhead box-O3 and activation of the AMPK pathway (Egan et al. 2011; Palikaras et al. 2015).

Other organs

Spinal cord ischemia–reperfusion (SCIR) injury is surgery-induced nervous system damage which occurs commonly after spinal cord, abdominal aortic and endovascular aortic repair surgeries. SCIR injury may contribute to neuronal degeneration, paraplegia, and quadriplegia. Various studies have suggested the oxidative stress, autophagy, inflammation, endothelial cell dysfunction and neuronal apoptosis are involved in SCIR injury as the underlying pathophysiological mechanisms (Beattie 2004). Lei et al. used an animal model of SCIR injury to investigate the effect of NAD⁺ on oxidative stress-induced neuronal apoptosis. They found that administration of NAD⁺ instantly after reperfusion decreased the oxidative stress level and neuronal apoptosis and improved the locomotors function (Xie et al. 2017a). Also, they showed that

NAD⁺ protects against SCIR injury-induced apoptosis via blocking autophagy (Xie et al. 2017b). NAMPT has been identified as a potent extracellular proinflammatory cytokine and can induce NF-κB pathway, leukocyte extravasation and toll-like receptor signaling (Sun et al. 2013). The FK866 reverted the cellular damages induced by spinal cord injury through preventing the elevation of inflammatory markers (TNF-α and IL-1β) (Esposito et al. 2012).

Nicotinamide has abrogated the acute lung injury (ALI) induced by I/R insult through the inhibition of PARP activity and suppression of production of nitric oxide, proinflammatory cytokines and free radicals with the restoration of ATP (Su et al. 2007). During acute I/R lung injury, the expression of NAMPT in the lungs is significantly increased in human and animal models, and the NAMPT levels are elevated in serum and bronchoalveolar lavage fluid. Inhibition of NAMPT ameliorated lung I/R injury through several mediators and signal pathways including decrease in the levels of inflammatory cytokines, reactive oxygen species, apoptosis, and NFκB and MAPK pathways (Wu et al. 2017). Similarly, Matsuda et al. demonstrated that inhibition of NAMPT could protect against intestinal I/R-associated ALI in mice via modulation of the NF-κB pathway (Matsuda et al. 2014). It seems that the function of NAMPT in different tissues under I/R injury is varied. Accumulating experiments have disclosed that activation of NAMPT provides cardiac and cerebral protection in I/R injury, whereas inhibition of NAMPT in the spinal cord, lung and intestinal tissues protect against I/R injury. Thus, more studies are needed to explore these contradictory findings.

Renal I/R injury often results from several clinical conditions including hypotension, sepsis, and surgical procedures (Dare et al. 2015). NMN can reduce blood urea nitrogen and serum creatinine levels and improve tubular damage in bilateral renal I/R injury in aged mice (Guan et al. 2017). Furthermore, nicotinamide adenine dinucleotide phosphate (NADPH) declined renal I/R injury and protected the kidneys from histological and functional damages. Additionally, NADPH decreased ROS production, blood urea nitrogen, creatinine levels, and inhibited apoptosis (Weng et al. 2018). The effects of NAD⁺ during I/R injury reported in studies have been summarized in Table 2.

Mitochondrial dysfunction in aging and ischemia and the role of NAD⁺

Mitochondria is the pivotal player in the regulation of apoptosis, cell viability, metabolism, calcium handling and energy homeostasis (Kuo et al. 2015; Palikaras and Tavernarakis 2014; Radogna et al. 2015). Mitochondria are dynamic organelles that continuously undergo fission and fusion. Mitochondrial fusion is associated with active mitochondria whereas fission happens among damaged mitochondria and contribute to their elimination by mitophagy, the specific autophagic removal of mitochondria (López-Lluch et al. 2008). Proteins mitofusin 1 and 2 (MFN1 and MFN2) and autosomal dominant optic atrophy-1 (OPA1) are involved in fusion, and dynamin related protein 1 (DRP1) and the fission protein 1 (FIS1) are involved in fission (Nakamura et al. 2006). These proteins regulate mitochondrial morphology and function. Mitochondrial dynamics contribute to several cellular and biological processes including proliferation, apoptosis (Liesa et al. 2009) and senescence (Hwang et al. 2009). Studies performed on aging models have revealed an imbalance in the activities of these mitochondrial dynamic proteins (Seo et al. 2010). During aging, mitochondria undergo morphological and functional changes as reduction of mitochondrial enzymes and proteins activities and impairment of its biogenesis (Yuan et al. 2016). These changes impair the oxidative-phosphorylation system, diminish the mitochondria turnover by inhibiting mitophagy and increase the accumulation of ROS and oxidation of lipids and proteins (Baldelli et al. 2014; Carelli et al. 2015; Ungvari et al. 2008).

Sirtuins are sensors of energy in cells and regulators of mitochondrial activity. They belong to a family of histone deacetylases that require NAD⁺ for the deacetylation reaction. The cellular energy levels modify sirtuins activity because of their dependence on NAD⁺ levels. In mammalian cells, seven sirtuins have been recognized (SIRT1–7). Silent information regulator 1 (SIRT1) and sirtuin 3 (SIRT3) are expressed in many tissues. SIRT1 modulates numerous physiological and pathological processes through its deacetylase activity. This enzyme mediates mitochondrial biogenesis through activation of peroxisome proliferator-activated receptor γ co-activator-1 α (PGC-1 α). The studies demonstrated that nuclear-mitochondrial communication disrupts during elderly

years because of decline in nuclear NAD⁺ levels and loss of SIRT1 activity (Gomes et al. 2013). The reduced activity of SIRT1 leads to a decrease in mitochondrial biogenesis, oxidative metabolism and antioxidant defense, resulting in damage to complex I of the electron transport chain and a decline in mitochondrial function (Imai and Guarente 2014). Thus, SIRT1 activation has protective effects against aging (Hernández-Jiménez et al. 2013). SIRT3 is located in the mitochondria and has been involved in regulation of autophagy by deacetylation of FOXO3 (Forkhead box protein O 3), modulation of the activity of mitochondrial manganese-superoxide dismutase and regulation of mitochondrial activity. SIRT3 influences longevity in humans. SIRT3 can also deacetylate cyclophilin D which is a regulator of mitochondrial permeability transition pore (mPTP), while mitochondrial NAD⁺ depletion during aging inhibits SIRT3 deacetylase activity and stimulates mPTP opening, consequently augments mitochondrial dysfunction (Hafner et al. 2010).

Mitochondrial dysfunction has been considered as one of the hallmarks of I/R injury. It has been induced by oxygen and glucose deprivation which occurs a few minutes after ischemia, leading to overproduction of ROS and depletion of ATP production. The increased generation of free radicals in the mitochondria and overloading calcium facilitate the opening of (mPTP) (Eltzschig and Eckle 2011; Shanmughapriya et al. 2015) which cause loss of membrane potential, increased free radical production and releasing of pro-apoptosis factors to cytoplasm (Zhou et al. 2017; Zorov et al. 2014). Also, mitochondrial dynamics disturb in I/R injury. The studies have reported that the expression of proteins Mfn2 and Opa1 are decreased in ischemic models, while the level of fission proteins, such as DRP1 and FIS increased, leading to mitochondrial dysfunction and disruption of Ca²⁺ homeostasis (Martorell-Riera et al. 2014; Peng et al. 2015).

Mitochondria are one of the key targets for treatment strategies in age-dependent diseases. They are critical for the preservation of normal cellular functions, primarily via regulation of energy production (Helley et al. 2017; Martin 2010). As noted above, a progressive reduction in mitochondrial function is the main hallmark of aging. One strategy for reversal of age-induced mitochondrial dysfunctions is increasing the intracellular levels of NAD⁺. In this regard, several studies have been done on elderly animal

Table 2 Summary of effects of NAMPT/NAD⁺ modulation in ischemic condition

NAD + – treatments	Dose/time	Species	Organ	Main findings	References
NAD ⁺	20 mg/kg	Rat	Heart	↓Myocardial ischemic injury and apoptotic damage, ↑antioxidant capacity	Zhang et al. (2016)
NMN	500 mg/kg	Mice	Heart	↑NAD ⁺ level, activation sirt1, stimulation autophagy flux	Yamamoto et al. (2014)
NAD ⁺	500,100 μm		H9c2 cardiac myoblasts	↓Apoptosis, ↑Sirt1 activity and ↓acetylation level of p53	Liu et al. (2014)
NR		Mice	Heart	↑Expression of PGC-1α, Tfam and NRF, ↓apoptosis and expression of LC3, P62, Beclin1, Atg5/7	Ping et al. (2016)
NAM	0.5 g/kg, 1 week	Mice	Heart	↑SUR2A mRNA and NAD ⁺ , ↓size of myocardial infarction	Sukhodub et al. (2010)
NAM	10 mmol/l	Rat	Cardiomyocyt	Improvement mitochondrial stress	Tong et al. (2012)
Overexpression of NAMPT		Mice	Heart	↑NAD ⁺ , ↓size of infract and apoptosis, stimulate autophagic flux	Becatti et al. (2012)
eNAMPT at the time of reperfusion			Heart	↓Myocardial injury by activation of PI3 K-Akt and MEK1/2-Erk1/2 and inhibition of mPTP opening	Lim et al. (2008)
eNAMPT			H9c2 myocardial cells	Inhibited apoptotic pathway, AMPK activation	Xiao et al. (2013)
NAD ⁺ after reperfusion	10 mg/kg		Brain	↓Neurological deficits and size infarct	Ying et al. (2007)
NAM	200 mg/kg	Rat	Brain	Improve remyelination	Wang et al. (2017)
NAD ⁺	50 mg/kg	Mice	Brain	↓Infarct size and edema formation, and neurological deficits by autophagy	Zheng et al. (2012)
NAMPT			Brain	Protects neurons against ischemic by SIRT1-dependent AMPK pathways	Wang et al. (2011)
NAM and NAD		Mice	Brain	↓Neuronal death and ↑mitochondrial biogenesis	Bi et al. (2012)
NAMPT over-expression			Brain	Promoted neurogenesis, survival rate and accelerates neurological function recovery and body weight gain	Zhao et al. (2015)
NMN	300 mg/kg	Mice	Brain	↓Apoptosis and infarction volume and release of TNF-α	Wei et al. (2017)
NAD ⁺	50 or 100 mg/kg	Rat	Spinal cord	↓Oxidative stress and neuronal apoptosis, improved locomotor function	Xie et al. (2017a)
NAD ⁺	75 mg/kg	Rat	Spinal cord	Protects against SCII via blocking autophagy	Xie et al. (2017b)
NMN	500 mg/kg	Mice	Renal	↓Blood urea nitrogen and serum creatinine levels and improved tubular damage	Dare et al. (2015)
NAM	100 mg/kg	Rat	Lung	↓Inflammation, ↑ATP	Su et al. (2007)

models to examine the effects of NR or NMN on mitochondrial functions in muscular tissues. They

have illustrated that NR or NMN administration enhance NAD⁺ mitochondrial levels, improve

mitochondrial function (Frederick et al. 2016; Mendelsohn and Larrick 2014; Mouchiroud et al. 2013) and activate SIRT1 and SIRT3 (Cantó et al. 2012). Williams et al. have demonstrated that Vitamin B3 supplement can prevent glaucoma in old mice via restoration of mitochondrial function (Williams et al. 2017).

In age-related neurodegenerative disorders such as Alzheimer, mitochondria are subjected to fragmentation (because of increased fission or decreased fusion events) and thus their morphology alters. In APP/PS1 Alzheimer mice treated with NMN, mitochondrial dynamics shifted towards fusion, leading to the amelioration of mitochondrial dysfunction (Long et al. 2015). Indeed, increasing the mitochondrial fusion proteins activity prevents mitochondrial membrane depolarization, cytochrome-c release, and consequently cell apoptosis (Breckenridge et al. 2003; Brooks et al. 2009). Also, NMN treatment has been shown to increase the levels of SIRT1 and PGC-1 α (Long et al. 2015).

Conclusions and future remarks

Increased elderly population is associated with the age-related diseases such as ischemic heart disease, neurological disorders such as stroke, and will increase the cost of health cares in coming decades. Aging is an independent risk factor for ischemia. NAD⁺ is involved in several cellular processes and it has been found that inadequate NAD⁺ is linked to a diversity of age-associated diseases. Recent studies have indicated that NAMPT–NAD⁺–SIRT1 axis plays pivotal roles in improving ischemic damages and delaying aging. Replenishment of NAD⁺ through NAD⁺ intermediates has anti-aging effects and can improve age-associated disorders and ischemic pathophysiology. Nevertheless, the major downstream mechanisms of NAD⁺ in different cellular compartments are not yet understood. Future studies are required to develop the NAD⁺ based therapies for improvement of aging-induced alterations and ischemia-related disorders including: (1) it is important to demarcate the accurate enzymes involved in the age-associated NAD⁺ reduction. (2) It is necessary to determine the effective dose of NAD⁺ in different diseases and aging. (3) The combination of CD38 inhibitors and NAD⁺ supplementations has not yet

been tested in treatment of age-related diseases and ischemic disorders. (4) Future studies are needed for analysis the NAD⁺-mediated signaling pathways in I/R insult and aging. (5) Clinical investigations are warranted on the effects of NAD⁺ administration in postponing aging phenotypes in humans. More investigations into the targets and functions of NAD⁺ will help to develop novel strategies for protection against I/R injury and other related diseases. NAD⁺ replenishment strategies may serve as a promising approach to delay senescence and improve the quality of life during elderly and combat with ischemic disorders and age-related pathologies.

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Compliance with ethical standards

Conflict of interest The authors declare that there are no conflicts of interest.

References

- Aksoy P, White TA, Thompson M, Chini EN (2006) Regulation of intracellular levels of NAD: a novel role for CD38. *Biochem Biophys Res Commun* 345:1386–1392
- Alano CC, Ying W, Swanson RA (2004) Poly (ADP-ribose) polymerase-1-mediated cell death in astrocytes requires NAD⁺ depletion and mitochondrial permeability transition. *J Biol Chem* 279:18895–18902
- Alano CC, Garnier P, Ying W, Higashi Y, Kauppinen TM, Swanson RA (2010) NAD⁺ Depletion Is Necessary and Sufficient for Poly (ADP-Ribose) Polymerase-1-Mediated Neuronal Death. *J Neurosci* 30:2967–2978
- Badalzadeh R, Azimi A, Alihemmati A, Yousefi B (2017) Chronic type-I diabetes could not impede the anti-inflammatory and anti-apoptotic effects of combined postconditioning with ischemia and cyclosporine A in myocardial reperfusion injury. *J Physiol Biochem* 73:111–120
- Baldelli S, Aquilano K, Ciriolo M (2014) PGC-1 α buffers ROS-mediated removal of mitochondria during myogenesis. *Cell Death Dis* 5:e1515
- Baxter P, Chen Y, Xu Y, Swanson RA (2014) Mitochondrial dysfunction induced by nuclear poly (ADP-ribose) polymerase-1: a treatable cause of cell death in stroke. *Trans Stroke Res* 5:136–144
- Bayrami G et al (2018) Combination of vildagliptin and ischemic postconditioning in diabetic hearts as a working strategy to reduce myocardial reperfusion injury by restoring mitochondrial function and autophagic activity. *Adv Pharm Bull* 8:319

- Beattie MS (2004) Inflammation and apoptosis: linked therapeutic targets in spinal cord injury. *Trends Mol Med* 10:580–583
- Becatti M, Taddei N, Cecchi C, Nassi N, Nassi PA, Fiorillo C (2012) SIRT1 modulates MAPK pathways in ischemic-reperfused cardiomyocytes. *Cell Mol Life Sci* 69:2245–2260
- Bi J, Li H, Ye SQ, Ding S (2012) Pre-B-cell colony-enhancing factor exerts a neuronal protection through its enzymatic activity and the reduction of mitochondrial dysfunction in in vitro ischemic models. *J Neurochem* 120:334–346
- Braidy N, Guillemin GJ, Mansour H, Chan-Ling T, Grant R (2011) Changes in kynurenine pathway metabolism in the brain, liver and kidney of aged female Wistar rats. *FEBS J* 278:4425–4434
- Breckenridge DG, Stojanovic M, Marcellus RC, Shore GC (2003) Caspase cleavage product of BAP31 induces mitochondrial fission through endoplasmic reticulum calcium signals, enhancing cytochrome c release to the cytosol. *J Cell Biol* 160:1115–1127
- Brooks C, Wei Q, Cho S-G, Dong Z (2009) Regulation of mitochondrial dynamics in acute kidney injury in cell culture and rodent models. *J Clin Investig* 119:1275–1285
- Brown KD et al (2014) Activation of SIRT3 by the NAD⁺ precursor nicotinamide riboside protects from noise-induced hearing loss. *Cell Metabol* 20:1059–1068
- Bueno M et al. (2018) Cyb5R3 an External Mitochondrial NADH-Dependent Reductase Confers Protection Against Lung Fibrosis. In: B107. MECHANISTIC ADVANCES IN PULMONARY FIBROSIS. American Thoracic Society, pp A4351–A4351
- Camacho-Pereira J et al (2016) CD38 dictates age-related NAD decline and mitochondrial dysfunction through an SIRT3-dependent mechanism. *Cell Metab* 23:1127–1139
- Cantó C et al (2012) The NAD⁺ precursor nicotinamide riboside enhances oxidative metabolism and protects against high-fat diet-induced obesity. *Cell Metab* 15:838–847
- Carelli V, Maresca A, Caporali L, Trifunov S, Zanna C, Rugolo M (2015) Mitochondria: biogenesis and mitophagy balance in segregation and clonal expansion of mitochondrial DNA mutations. *Int J Biochem Cell Biol* 63:21–24
- Chen S-D, Yang D-I, Lin T-K, Shaw F-Z, Liou C-W, Chuang Y-C (2011) Roles of oxidative stress, apoptosis, PGC-1 α and mitochondrial biogenesis in cerebral ischemia. *Int J Mol Sci* 12:7199–7215
- Chini EN (2009) CD38 as a regulator of cellular NAD: a novel potential pharmacological target for metabolic conditions. *Curr Pharm Des* 15:57–63
- Chini CC, Guerrico AMG, Nin V, Camacho-Pereira J, Escande C, Barbosa MT, Chini EN (2014) Targeting of NAD metabolism in pancreatic cancer cells: potential novel therapy for pancreatic tumors. *Clin Cancer Res* 20:120–130
- Choe CU et al (2011) CD38 exacerbates focal cytokine production, postischemic inflammation and brain injury after focal cerebral ischemia. *PLoS ONE* 6:e19046
- Claire C et al (2003) CD38-dependent ADP-ribosyl cyclase activity in developing and adult mouse brain. *Biochem J* 370:175–183
- D'Annunzio V, Perez V, Boveris A, Gelpi RJ, Poderoso JJ (2016) Role of thioredoxin-1 in ischemic preconditioning, postconditioning and aged ischemic hearts. *Pharmacol Res* 109:24–31
- Dare AJ, Bolton EA, Pettigrew GJ, Bradley JA, Saeb-Parsy K, Murphy MP (2015) Protection against renal ischemia-reperfusion injury in vivo by the mitochondria targeted antioxidant. *MitoQ Redox Biol* 5:163–168
- de Picciotto NE et al (2016) Nicotinamide mononucleotide supplementation reverses vascular dysfunction and oxidative stress with aging in mice. *Aging Cell* 15:522–530
- Di Lisa F, Menabò R, Canton M, Barile M, Bernardi P (2000) Opening of the mitochondrial permeability transition pore causes depletion of mitochondrial and cytosolic NAD⁺ and is a causative event in the death of myocytes in post-ischemic reperfusion of the heart. *J Biol Chem* 276(4):2571–2575
- Egan DF et al (2011) Phosphorylation of ULK1 (hATG1) by AMP-activated protein kinase connects energy sensing to mitophagy. *Science* 331:456–461
- Elliott G, Rechsteiner M (1975) Pyridine nucleotide metabolism in mitotic cells. *J Cell Physiol* 86:641–651
- Eltzschig HK, Eckle T (2011) Ischemia and reperfusion—from mechanism to translation. *Nat Med* 17:1391
- Esposito E et al (2012) The NAMPT inhibitor FK866 reverts the damage in spinal cord injury. *J Neuroinflamm* 9:66
- Fang EF et al (2014) Defective mitophagy in XPA via PARP-1 hyperactivation and NAD⁺/SIRT1 reduction. *Cell* 157:882–896
- Frederick DW et al (2016) Loss of NAD homeostasis leads to progressive and reversible degeneration of skeletal muscle. *Cell Metab* 24:269–282
- Gems D, Partridge L (2013) Genetics of longevity in model organisms: debates and paradigm shifts. *Annu Rev Physiol* 75:621–644
- Gomes AP et al (2013) Declining NAD⁺ induces a pseudo-hypoxic state disrupting nuclear-mitochondrial communication during aging. *Cell* 155:1624–1638
- Guan X-H et al (2016) CD38 deficiency protects the heart from ischemia/reperfusion injury through activating SIRT1/FOXOs-mediated antioxidative stress pathway. *Oxid Med cell Longev*. <https://doi.org/10.1155/2016/7410257>
- Guan Y, Wang S-R, Huang X-Z, Xie Q-h XuY-Y, Shang D, Hao C-M (2017) Nicotinamide Mononucleotide, an NAD⁺ Precursor, Rescues Age-Associated Susceptibility to AKI in a Sirtuin 1-Dependent Manner. *J Am Soc Nephrol* 28(8):2337–2352
- Hafner AV, Dai J, Gomes AP, Xiao C-Y, Palmeira CM, Rosenzweig A, Sinclair DA (2010) Regulation of the mPTP by SIRT3-mediated deacetylation of CypD at lysine 166 suppresses age-related cardiac hypertrophy. *Aging (Albany NY)* 2:914
- Han J, Shi S, Min L, Wu T, Xia W, Ying W (2011) NAD⁺ treatment induces delayed autophagy in Neuro2a cells partially by increasing oxidative stress. *Neurochem Res* 36:2270
- Harden A, Young WJ (1906) The alcoholic ferment of yeast-juice Part II.—The coferment of yeast-juice. *Proc R Soc Lond B* 78:369–375

- Harman D (1956) A theory based on free radical and radiation chemistry. *J. Biol. Chem.* 219:305–314
- Helley MP, Pinnell J, Sportelli C, Tieu K (2017) Mitochondria: a common target for genetic mutations and environmental toxicants in parkinson's disease. *Front Genet* 8:177
- Hernández-Jiménez M et al (2013) Silent information regulator 1 protects the brain against cerebral ischemic damage. *Stroke* 44:2333–2337
- Houtkooper RH, Auwerx J (2012) Exploring the therapeutic space around NAD⁺. Rockefeller University Press, New York
- Hsu C-P, Oka S, Shao D, Hariharan N, Sadoshima J (2009) Nicotinamide phosphoribosyltransferase regulates cell survival through NAD⁺ synthesis in cardiac myocytes. *Circ Res* 105:481–491
- Hwang ES, Yoon G, Kang HT (2009) A comparative analysis of the cell biology of senescence and aging. *Cell Mol Life Sci* 66:2503–2524
- Imai SI, Guarente L (2014) NAD⁺ and sirtuins in aging and disease. *Trends Cell Biol* 24:464–471
- Imai SI, Yoshino J (2013) The importance of NAMPT/NAD⁺/SIRT1 in the systemic regulation of metabolism and ageing. *Diabetes. Obesity Metab* 15:26–33
- Jang SY, Kang HT, Hwang ES (2012) Nicotinamide-induced mitophagy: an event mediated by high NAD⁺/NADH ratio and SIRT1 activation. *J Biol Chem* 287(23):19304
- Khalili H, Talasaz AH, Jenab Y, Salarifar M (2012) Clinical characteristics and risk assessment of ST-segment elevation myocardial infarction patients of an Iranian referral center. *J Cardiovasc Med* 13:708–715
- Kitani T, Okuno S, Fujisawa H (2003) Growth phase-dependent changes in the subcellular localization of pre-B-cell colony-enhancing factor 1. *FEBS Lett* 544:74–78
- Kuo C-Y, Chiu Y-C, Lee AY-L, Hwang T-L (2015) Mitochondrial Lon protease controls ROS-dependent apoptosis in cardiomyocyte under hypoxia. *Mitochondrion* 23:7–16
- Lee JS et al (2012) Beta-lapachone, a modulator of NAD metabolism, prevents health declines in aged mice. *PLoS ONE* 7:e47122
- Lee CF et al (2016) Normalization of NAD⁺ redox balance as a therapy for heart failure. *Circulation* 134(12):883
- Levine ME et al (2014) Low protein intake is associated with a major reduction in IGF-1, cancer, and overall mortality in the 65 and younger but not older population. *Cell Metab* 19:407–417
- Li J et al (2012) Reperfusion promotes mitochondrial dysfunction following focal cerebral ischemia in rats. *PLoS ONE* 7:e46498
- Liesa M, Palacín M, Zorzano A (2009) Mitochondrial dynamics in mammalian health and disease. *Physiol Rev* 89:799–845
- Lim SY, Davidson SM, Paramanathan AJ, Smith CC, Yellon DM, Hausenloy DJ (2008) The novel adipocytokine visfatin exerts direct cardioprotective effects. *J Cell Mol Med* 12:1395–1403
- Liu JP (2014) Molecular mechanisms of ageing and related diseases. *Clin Exp Pharmacol Physiol* 41:445–458
- Liu D, Pitta M, Mattson MP (2008) Preventing NAD⁺ depletion protects neurons against excitotoxicity: bioenergetic effects of mild mitochondrial uncoupling and caloric restriction. *Ann N Y Acad Sci* 1147:275–282
- Liu L, Wang P, Liu X, He D, Liang C, Yu Y (2014) Exogenous NAD⁺ supplementation protects H9c2 cardiac myoblasts against hypoxia/reoxygenation injury via Sirt1-p53 pathway. *Fundam Clin Pharmacol* 28:180–189
- Long AN, Owens K, Schlappal AE, Kristian T, Fishman PS, Schuh RA (2015) Effect of nicotinamide mononucleotide on brain mitochondrial respiratory deficits in an Alzheimer's disease-relevant murine model. *BMC Neurol* 15:19
- Long A, Park JH, Klimova N, Fowler C, Loane DJ, Kristian T (2017) CD38 knockout mice show significant protection against ischemic brain damage despite high level poly-ADP-ribosylation. *Neurochem Res* 42:283–293
- López-Lluch G, Irusta PM, Navas P, de Cabo R (2008) Mitochondrial biogenesis and healthy aging. *Exp Gerontol* 43:813–819
- López-Otín C, Blasco MA, Partridge L, Serrano M, Kroemer G (2013) The hallmarks of aging. *Cell* 153:1194–1217
- Lu L-F et al (2009) Elevated visfatin/pre-B-cell colony-enhancing factor plasma concentration in ischemic stroke. *J Stroke Cerebrovasc Dis* 18:354–359
- Martin LJ (2010) Mitochondrial and cell death mechanisms in neurodegenerative diseases. *Pharmacol Ther* 3:839–915
- Martorell-Riera A et al (2014) Mfn2 downregulation in excitotoxicity causes mitochondrial dysfunction and delayed neuronal death. *EMBO J* 33:2388–2407
- Matsuda A et al (2014) FK866, a visfatin inhibitor, protects against acute lung injury after intestinal ischemia-reperfusion in mice via NF- κ B pathway. *Ann Surg* 259:1007–1017
- Matsuura S, Egi Y, Yuki S, Horikawa T, Satoh H, Akira T (2011) MP-124, a novel poly (ADP-ribose) polymerase-1 (PARP-1) inhibitor, ameliorates ischemic brain damage in a non-human primate model. *Brain Res* 1410:122–131
- Mendelsohn AR, Larrick JW (2014) Partial reversal of skeletal muscle aging by restoration of normal NAD⁺ levels. *Rejuvenation Res* 17:62–69
- Mills KF et al (2016) Long-term administration of nicotinamide mononucleotide mitigates age-associated physiological decline in mice. *Cell Metab* 24:795–806
- Miquel J, Economos A, Fleming J, Johnson J Jr (1980) Mitochondrial role in cell aging. *Exp Gerontol* 15:575–591
- Mokudai T, Ayoub IA, Sakakibara Y, Lee E-J, Ogilvy CS, Maynard KI (2000) Delayed treatment with nicotinamide (vitamin B3) improves neurological outcome and reduces infarct volume after transient focal cerebral ischemia in Wistar rats. *Stroke* 31:1679–1685
- Moraga A et al (2015) Aging increases microglial proliferation, delays cell migration, and decreases cortical neurogenesis after focal cerebral ischemia. *J Neuroinflamm* 12:87
- Mouchiroud L et al (2013) The NAD⁺/sirtuin pathway modulates longevity through activation of mitochondrial UPR and FOXO signaling. *Cell* 154:430–441
- Murphy E, Steenbergen C (2008) Mechanisms underlying acute protection from cardiac ischemia-reperfusion injury. *Physiol Rev* 88:581–609
- Nakagawa T, Guarente L (2014) SnapShot: sirtuins, NAD⁺, and aging. *Cell Metab* 20:192
- Nakamura N, Kimura Y, Tokuda M, Honda S, Hirose S (2006) MARCH-V is a novel mitofusin 2-and Drp1-binding protein able to change mitochondrial morphology. *EMBO Rep* 7:1019–1022

- Palikaras K, Tavernarakis N (2014) Mitochondrial homeostasis: the interplay between mitophagy and mitochondrial biogenesis. *Exp Gerontol* 56:182–188
- Palikaras K, Lionaki E, Tavernarakis N (2015) Coordination of mitophagy and mitochondrial biogenesis during ageing in *C. elegans*. *Nature* 521:525
- Park JH, Long A, Owens K, Kristian T (2016) Nicotinamide mononucleotide inhibits post-ischemic NAD⁺ degradation and dramatically ameliorates brain damage following global cerebral ischemia. *Neurobiol Dis* 95:102–110
- Peng C et al (2015) Mitofusin 2 ameliorates hypoxia-induced apoptosis via mitochondrial function and signaling pathways. *Int J Biochem Cell Biol* 69:29–40
- Pillai VB, Sundaresan NR, Kim G, Samant S, Moreno-Vinasco L, Garcia JG, Gupta MP (2012) Namp1 secreted from cardiomyocytes promotes development of cardiac hypertrophy and adverse ventricular remodeling. *Am J Physiol-Heart Circ Physiol* 304:H415–H426
- Ping L, Yin Z, Li Y (2016) GW27-e0632 NR reduced myocardial ischemia-reperfusion injury by improving mitochondrial biogenesis and reducing excessive autophagy via Sirt3-PGC-1 α /P53 pathway. *J Am Coll Cardiol* 68:C24–C25
- Polzonetti V, Carpi FM, Micozzi D, Pucciarelli S, Vincenzetti S, Napolioni V (2012) Population variability in CD38 activity: correlation with age and significant effect of TNF- α -308G > A and CD38 184C > G SNPs. *Mol Genet Metab* 105:502–507
- Radogna F, Albertini M, De Nicola M, Diederich M, Bejarano I, Ghibelli L (2015) Melatonin promotes Bax sequestration to mitochondria reducing cell susceptibility to apoptosis via the lipoxygenase metabolite 5-hydroxyeicosatetraenoic acid. *Mitochondrion* 21:113–121
- Rechsteiner M, Hillyard D, Olivera BM (1976) Turnover of nicotinamide adenine dinucleotide in cultures of human cells. *J Cell Physiol* 88:207–217
- Revollo JR et al (2007) Namp1/PBEF/visfatin regulates insulin secretion in β cells as a systemic NAD biosynthetic enzyme. *Cell Metab* 6:363–375
- Rex A, Spychalla M, Fink H (2004) Treatment with reduced nicotinamide adenine dinucleotide (NADH) improves water maze performance in old Wistar rats. *Behav Brain Res* 154:149–153
- Rodgers JT, Lerin C, Gerhart-Hines Z, Puigserver P (2008) Metabolic adaptations through the PGC-1 α and SIRT1 pathways. *FEBS Lett* 582:46–53
- Romero JR (2007) Prevention of ischemic stroke: overview of traditional risk factors. *Curr Drug Targets* 8:794–801
- Saeid F, Aniseh J, Reza B, Manouchehr VS (2018) Signaling mediators modulated by cardioprotective interventions in healthy and diabetic myocardium with ischaemia-reperfusion injury. *Eur J Prev Cardiol* 25(14):1463
- Salehpour F et al (2019) Photobiomodulation and Coenzyme Q10 Treatments Attenuate Cognitive Impairment Associated with Model of Transient Global Brain Ischemia in Artificially Aged Mice. *Front Cell Neurosci* 13:74
- Samal B, Sun Y, Stearns G, Xie C, Suggs S, McNiece I (1994) Cloning and characterization of the cDNA encoding a novel human pre-B-cell colony-enhancing factor. *Mol Cell Biol* 14:1431–1437
- Scheibye-Knudsen M et al (2014) A high-fat diet and NAD⁺ activate Sirt1 to rescue premature aging in cockayne syndrome. *Cell Metab* 20:840–855
- Seo AY, Joseph A-M, Dutta D, Hwang JC, Aris JP, Leeuwenburgh C (2010) New insights into the role of mitochondria in aging: mitochondrial dynamics and more. *J Cell Sci* 123:2533–2542
- Shanmughapriya S et al (2015) SPG7 is an essential and conserved component of the mitochondrial permeability transition pore. *Mol Cell* 60:47–62
- Smith CM, Chen Y, Sullivan ML, Kochanek PM, Clark RS (2011) Autophagy in acute brain injury: feast, famine, or folly? *Neurobiol Dis* 43:52–59
- Song HK et al (2008) Visfatin: a new player in mesangial cell physiology and diabetic nephropathy. *Am J Physiol-Renal Physiol* 295:F1485–F1494
- Song L et al (2014) Inhibition of 12/15 lipoxygenase by baicalin reduces myocardial ischemia/reperfusion injury via modulation of multiple signaling pathways. *Apoptosis* 19:567–580
- Stein LR, Imai SI (2014) Specific ablation of Namp1 in adult neural stem cells recapitulates their functional defects during aging. *EMBO J* 33:1321–1340
- Strosznajder RP, Gadamski R, Czapski GA, Jesko H, Strosznajder JB (2003) Poly (ADP-ribose) polymerase during reperfusion after transient forebrain ischemia. *J Mol Neurosci* 20:61–71
- Su C-F, Liu DD, Kao SJ, Chen HI (2007) Nicotinamide abrogates acute lung injury caused by ischemia-reperfusion. *Eur Respir J* 30(2):199
- Sukhodub A, Du Q, Jovanović S, Jovanović A (2010) Nicotinamide-rich diet protects the heart against ischaemia-reperfusion in mice: a crucial role for cardiac SUR2A. *Pharmacol Res* 61:564–570
- Sun Z, Lei H, Zhang Z (2013) Pre-B cell colony enhancing factor (PBEF), a cytokine with multiple physiological functions. *Cytokine Growth Factor Rev* 24:433–442
- Tong D-L et al (2012) Nicotinamide pretreatment protects cardiomyocytes against hypoxia-induced cell death by improving mitochondrial stress. *Pharmacology* 90:11–18
- Ungvari Z, Labinskyy N, Gupte S, Chander PN, Edwards JG, Csizsar A (2008) Dysregulation of mitochondrial biogenesis in vascular endothelial and smooth muscle cells of aged rats. *Am J Physiol-Heart Circ Physiol* 294:H2121–H2128
- Van Gool F et al (2009) Intracellular NAD levels regulate tumor necrosis factor protein synthesis in a sirtuin-dependent manner. *Nat Med* 15:206
- Veith S, Mangerich A (2015) RecQ helicases and PARP1 team up in maintaining genome integrity. *Ageing Res Rev* 23:12–28
- von Euler H (1999) Fermentation of sugars and fermentative enzymes, In: Nobelstiftelsen (eds) Nobel Lectures in Chemistry, New Jersey: World Scientific, pp. 144–155
- Wallace DC (2012) Mitochondria and cancer. *Nat Rev Cancer* 12:685
- Wang P et al (2011) Nicotinamide phosphoribosyltransferase protects against ischemic stroke through SIRT1-dependent adenosine monophosphate-activated kinase pathway. *Ann Neurol* 69:360–374

- Wang P, Guan Y-F, Du H, Zhai Q-W, Su D-F, Miao C-Y (2012) Induction of autophagy contributes to the neuroprotection of nicotinamide phosphoribosyltransferase in cerebral ischemia. *Autophagy* 8:77–87
- Wang C, Zhang Y, Ding J, Zhao Z, Qian C, Luan Y, Teng G-J (2017) Nicotinamide administration improves remyelination after stroke. *Neural Plast* 2017:12
- Wei CC et al (2017) NAD replenishment with nicotinamide mononucleotide protects blood–brain barrier integrity and attenuates delayed tissue plasminogen activator-induced haemorrhagic transformation after cerebral ischaemia. *Br J Pharmacol* 174:3823–3836
- Weng XF, Li ST, Song Q, Zhu Q, D-d Song, Qin ZH, Xie Y (2018) Protective effect of nicotinamide adenine dinucleotide phosphate on renal ischemia-reperfusion injury. *Kidney Blood Press Res* 43:651–663
- Williams PA et al (2017) Vitamin B3 modulates mitochondrial vulnerability and prevents glaucoma in aged mice. *Science* 355:756–760
- Wu G-C et al (2017) Targeting of nicotinamide phosphoribosyltransferase enzymatic activity ameliorates lung damage induced by ischemia/reperfusion in rats. *Respir Res* 18:71
- Xiao J, Sun B, Li M, Wu Y, Sun XB (2013) A novel adipocytokine visfatin protects against H₂O₂-induced myocardial apoptosis: a missing link between obesity and cardiovascular disease. *J Cell Physiol* 228:495–501
- Xie L, Wang Z, Li C, Yang K, Liang Y (2017a) Protective effect of nicotinamide adenine dinucleotide (NAD⁺) against spinal cord ischemia–reperfusion injury via reducing oxidative stress-induced neuronal apoptosis. *J Clin Neurosci* 36:114–119
- Xie L, Yu S, Wang Z, Yang K, Liu Z, Li C, Liang Y (2017b) Nicotinamide adenine dinucleotide protects against spinal cord ischemia reperfusion injury-induced apoptosis by blocking autophagy. *Oxid Med Cell Longev*. <https://doi.org/10.1155/2017/7063874>
- Yamamoto T, Byun J, Zhai P, Ikeda Y, Oka S, Sadoshima J (2014) Nicotinamide mononucleotide, an intermediate of NAD⁺ synthesis, protects the heart from ischemia and reperfusion. *PLoS ONE* 9:e98972
- Yang H et al (2007) Nutrient-sensitive mitochondrial NAD⁺ levels dictate cell survival. *Cell* 130:1095–1107
- Ying W (2008) NAD⁺/NADH and NADP⁺/NADPH in cellular functions and cell death: regulation and biological consequences. *Antioxid Redox Signal* 10:179–206
- Ying W et al (2007) Intranasal administration with NAD⁺ profoundly decreases brain injury in a rat model of transient focal ischemia. *Front Biosci* 12:2728–2734
- Yoshino J, Mills KF, Yoon MJ, S-i Imai (2011) Nicotinamide mononucleotide, a key NAD⁺ intermediate, treats the pathophysiology of diet- and age-induced diabetes in mice. *Cell Metab* 14:528–536
- Yuan Y, Cruzat VF, Newsholme P, Cheng J, Chen Y, Lu Y (2016) Regulation of SIRT1 in aging: roles in mitochondrial function and biogenesis. *Mech Ageing Dev* 155:10–21
- Zhang T et al (2009) Enzymes in the NAD⁺ salvage pathway regulate SIRT1 activity at target gene promoters. *J Biol Chem* 284(30):20408
- Zhang Y et al (2016) Exogenous NAD⁺ administration significantly protects against myocardial ischemia/reperfusion injury in rat model. *Am J Trans Res* 8:3342
- Zhao Y, Liu XZ, Tian WW, Guan YF, Wang P, Miao CY (2014) Extracellular visfatin has nicotinamide phosphoribosyltransferase enzymatic activity and is neuroprotective against ischemic injury. *CNS Neurosci Ther* 20:539–547
- Zhao Y et al (2015) Regenerative neurogenesis after ischemic stroke promoted by nicotinamide phosphoribosyltransferase–nicotinamide adenine dinucleotide cascade. *Stroke* 46:1966–1974
- Zheng C, Han J, Xia W, Shi S, Liu J, Ying W (2012) NAD⁺ administration decreases ischemic brain damage partially by blocking autophagy in a mouse model of brain ischemia. *Neurosci Lett* 512:67–71
- Zhou H et al (2017) Ripk3 induces mitochondrial apoptosis via inhibition of FUNDC1 mitophagy in cardiac IR injury. *Redox Biol* 13:498–507
- Zorov DB, Juhaszova M, Sollott SJ (2014) Mitochondrial reactive oxygen species (ROS) and ROS-induced ROS release. *Physiol Rev* 94:909–950

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